

```
In [1]: import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt
```

```
In [31]: df=pd.read_csv('StudentsPerformance.csv')
df
```

Out[31]:

	gender	race/ethnicity	parental level of education	lunch	test preparation course	math score	reading score	writing score
0	female	group B	bachelor's degree	standard	none	72	72	74
1	female	group C	some college	standard	completed	69	90	88
2	female	group B	master's degree	standard	none	90	95	93
3	male	group A	associate's degree	free/reduced	none	47	57	44
4	male	group C	some college	standard	none	76	78	75
...	...	...	...	...	...	...	...	...
995	female	group E	master's degree	standard	completed	88	99	95
996	male	group C	high school	free/reduced	none	62	55	55
997	female	group C	high school	free/reduced	completed	59	71	65
998	female	group D	some college	standard	completed	68	78	77
999	female	group D	some college	free/reduced	none	77	86	86

1000 rows × 8 columns

```
In [7]: df.head()
```

Out[7]:

	gender	race/ethnicity	parental level of education	lunch	test preparation course	math score	reading score	writing score
0	female	group B	bachelor's degree	standard	none	72	72	74
1	female	group C	some college	standard	completed	69	90	88
2	female	group B	master's degree	standard	none	90	95	93
3	male	group A	associate's degree	free/reduced	none	47	57	44
4	male	group C	some college	standard	none	76	78	75

```
In [9]: df.tail()
```

Out[9]:

	gender	race/ethnicity	parental level of education	lunch	test preparation course	math score	reading score	writing score
995	female	group E	master's degree	standard	completed	88	99	95
996	male	group C	high school	free/reduced	none	62	55	55
997	female	group C	high school	free/reduced	completed	59	71	65
998	female	group D	some college	standard	completed	68	78	77
999	female	group D	some college	free/reduced	none	77	86	86

```
In [11]: df.describe()
```

Out[11]:

	math score	reading score	writing score
count	1000.00000	1000.000000	1000.000000
mean	66.08900	69.169000	68.054000
std	15.16308	14.600192	15.195657
min	0.00000	17.000000	10.000000
25%	57.00000	59.000000	57.750000
50%	66.00000	70.000000	69.000000
75%	77.00000	79.000000	79.000000
max	100.00000	100.000000	100.000000

```
In [13]: df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
```

```
RangeIndex: 1000 entries, 0 to 999
```

```
Data columns (total 8 columns):
```

#	Column	Non-Null Count	Dtype
0	gender	1000 non-null	object
1	race/ethnicity	1000 non-null	object
2	parental level of education	1000 non-null	object
3	lunch	1000 non-null	object
4	test preparation course	1000 non-null	object
5	math score	1000 non-null	int64
6	reading score	1000 non-null	int64
7	writing score	1000 non-null	int64

```
dtypes: int64(3), object(5)
```

```
memory usage: 62.6+ KB
```

```
In [15]: df['test preparation course'].value_counts(normalize=True)*100
```

```
Out[15]: test preparation course
none      64.2
completed 35.8
Name: proportion, dtype: float64
```

```
In [17]: df['race/ethnicity'].value_counts(normalize=True)*100
```

```
Out[17]: race/ethnicity
group C    31.9
group D    26.2
group B    19.0
group E    14.0
group A     8.9
Name: proportion, dtype: float64
```

```
In [19]: df['gender'].value_counts(normalize=True)*100
```

```
Out[19]: gender
female    51.8
male      48.2
Name: proportion, dtype: float64
```

```
In [21]: df['lunch'].value_counts(normalize=True)*100
```

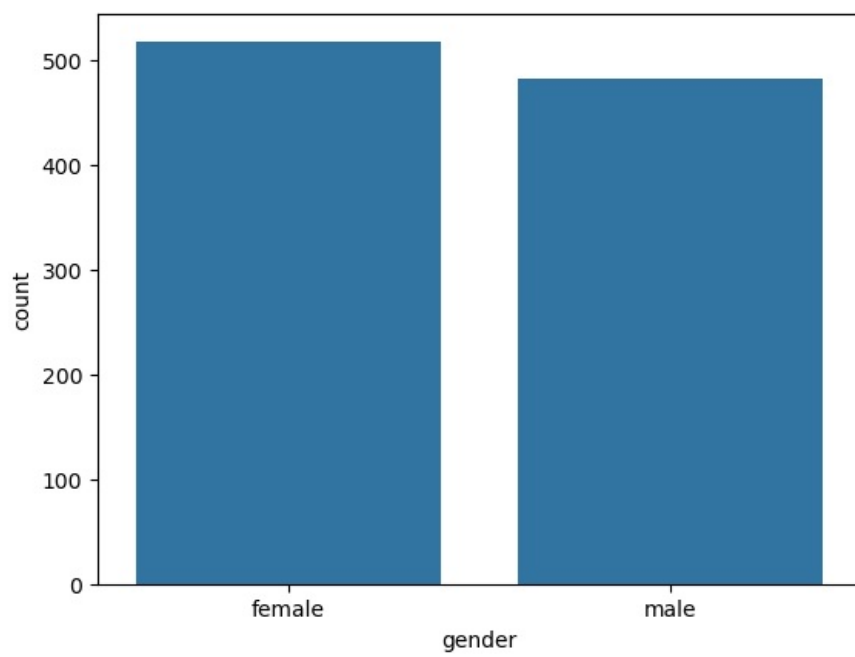
```
Out[21]: lunch
standard    64.5
free/reduced 35.5
Name: proportion, dtype: float64
```

```
In [25]: df['parental level of education'].value_counts(normalize=True)*100
```

```
Out[25]: parental level of education
some college    22.6
associate's degree 22.2
high school    19.6
some high school 17.9
bachelor's degree 11.8
master's degree  5.9
Name: proportion, dtype: float64
```

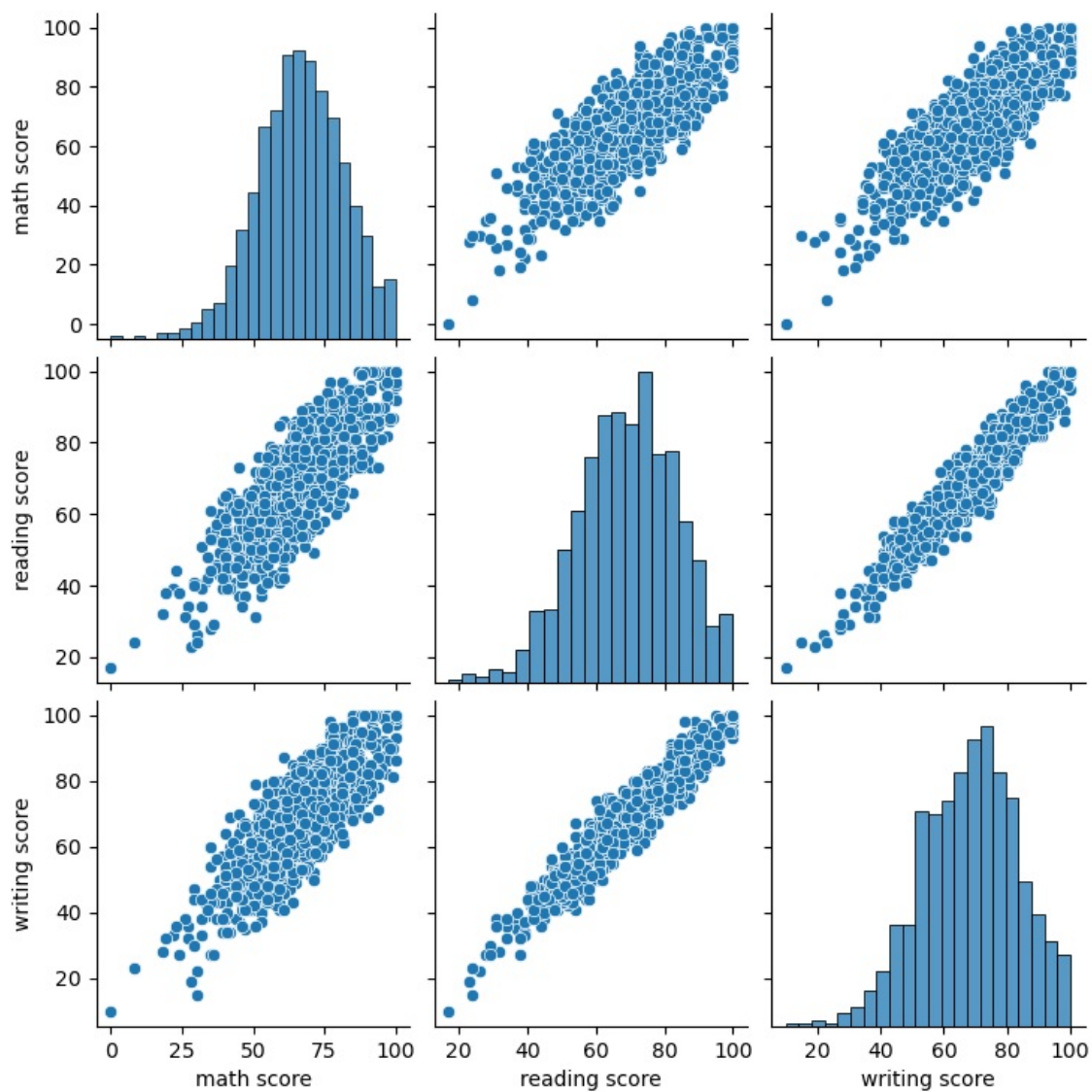
```
In [35]: sns.countplot(df,x='gender')
```

```
Out[35]: <Axes: xlabel='gender', ylabel='count'>
```



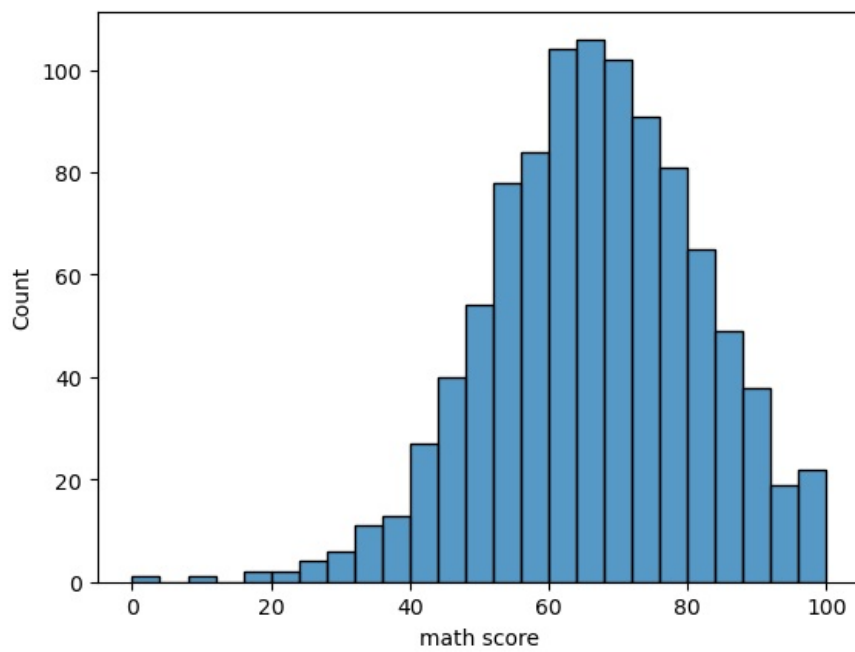
```
In [37]: sns.pairplot(df)
```

```
Out[37]: <seaborn.axisgrid.PairGrid at 0x1f16cdd49e0>
```



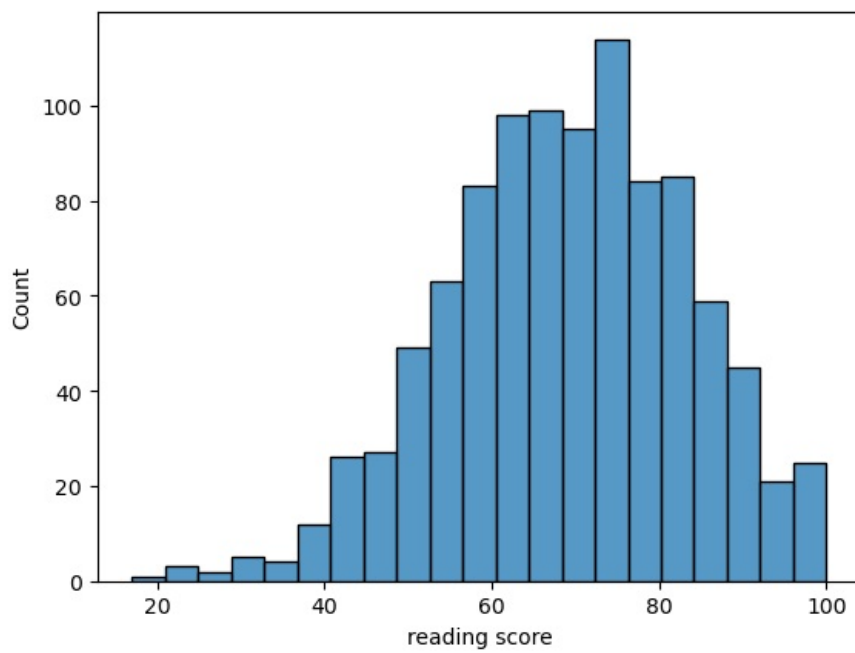
```
In [45]: sns.histplot(df,x='math score')
```

```
Out[45]: <Axes: xlabel='math score', ylabel='Count'>
```



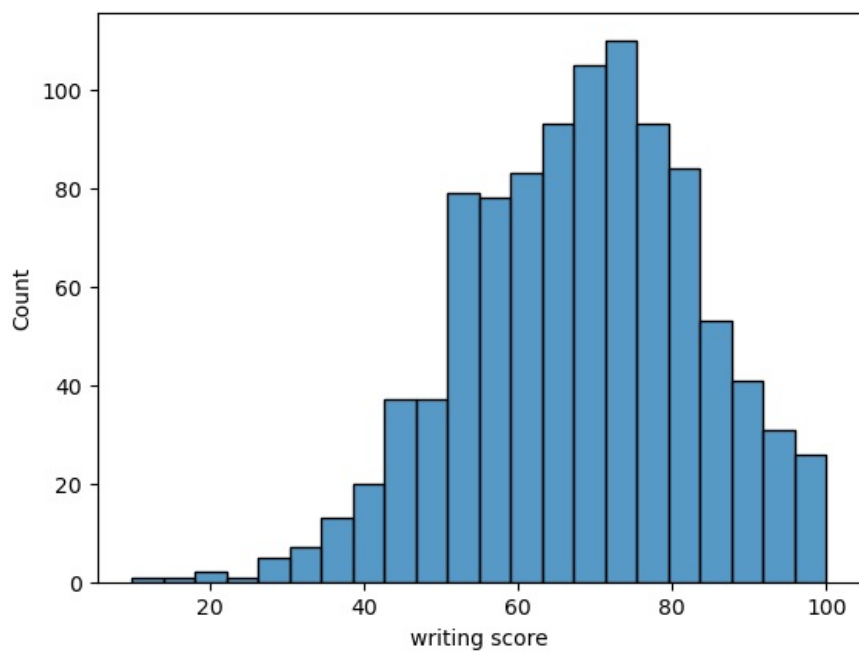
```
In [47]: sns.histplot(df,x='reading score')
```

```
Out[47]: <Axes: xlabel='reading score', ylabel='Count'>
```



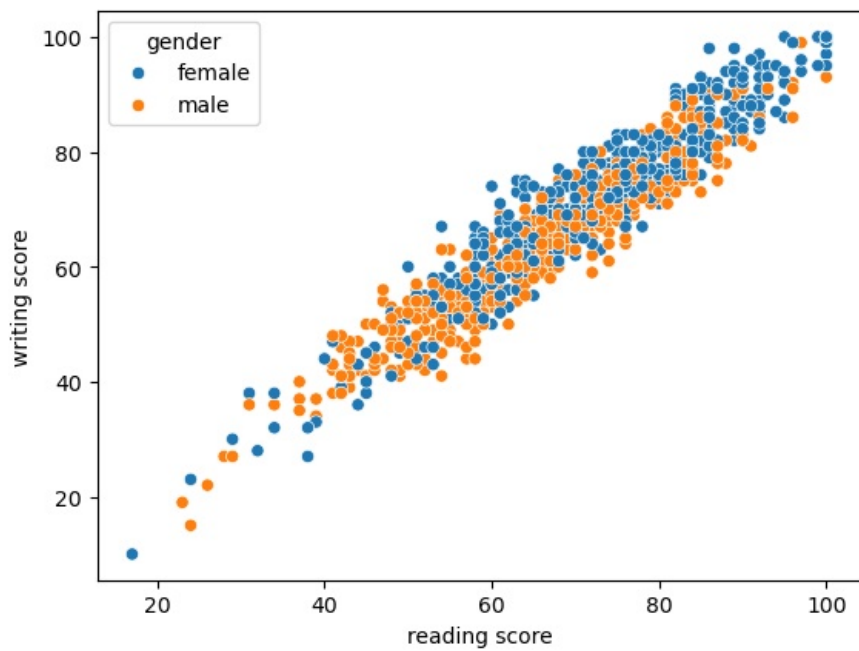
```
In [49]: sns.histplot(df,x='writing score')
```

```
Out[49]: <Axes: xlabel='writing score', ylabel='Count'>
```



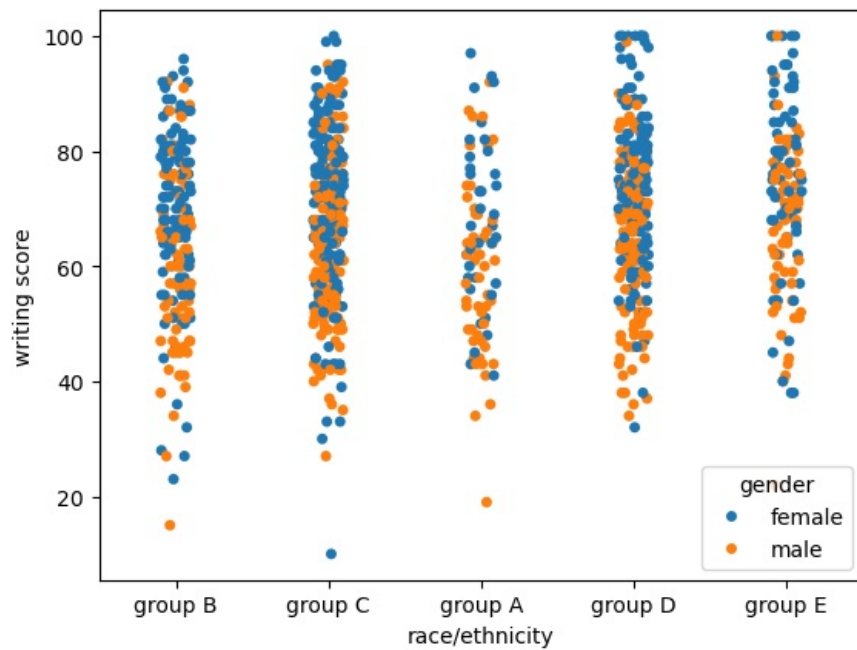
```
In [51]: sns.scatterplot(df,x='reading score',y='writing score',hue='gender')
```

```
Out[51]: <Axes: xlabel='reading score', ylabel='writing score'>
```



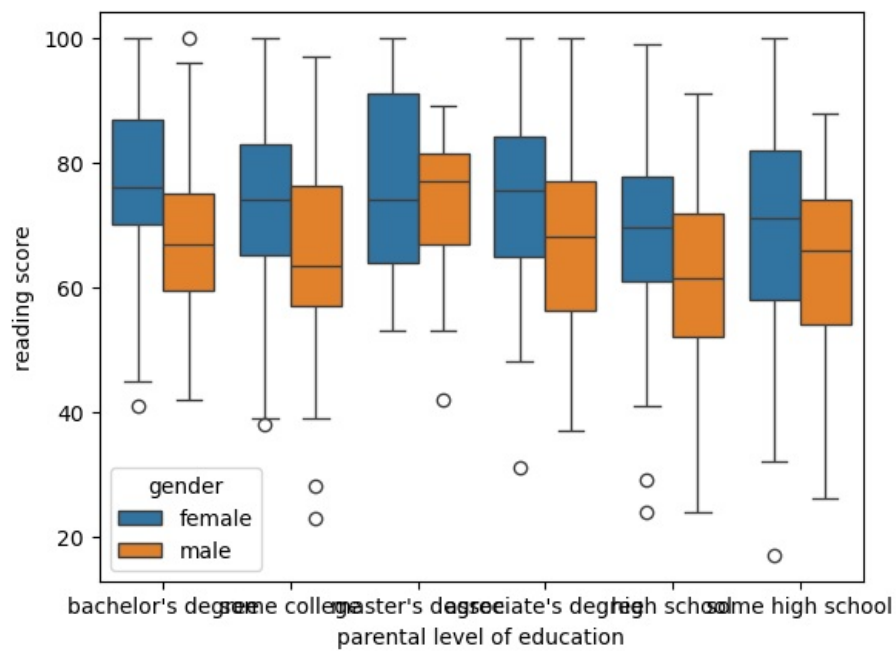
```
In [53]: sns.stripplot(df,x='race/ethnicity',y='writing score',hue='gender')
```

```
Out[53]: <Axes: xlabel='race/ethnicity', ylabel='writing score'>
```



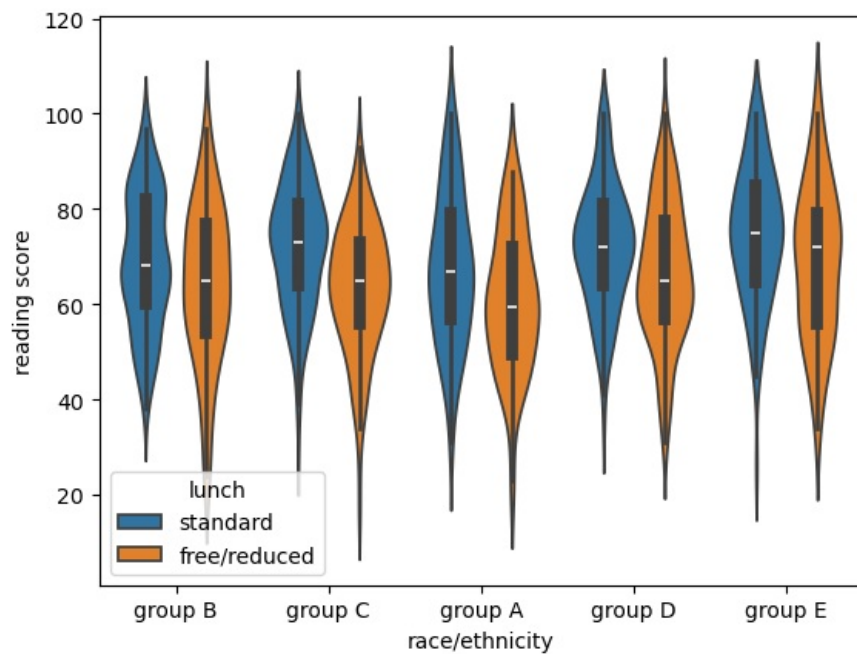
```
In [61]: sns.boxplot(df,x='parental level of education',y='reading score',hue='gender')
```

```
Out[61]: <Axes: xlabel='parental level of education', ylabel='reading score'>
```



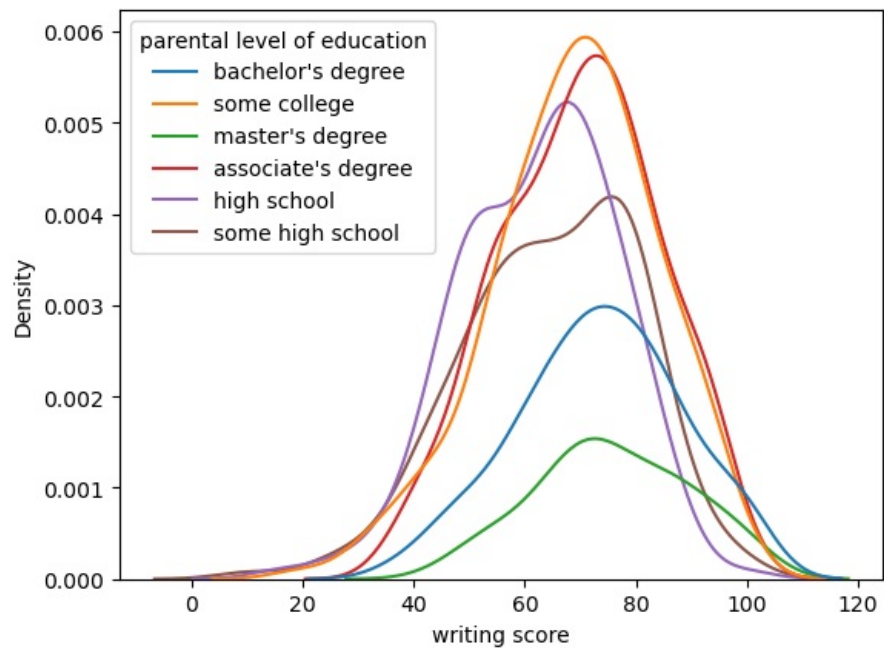
```
In [63]: sns.violinplot(df,x='race/ethnicity',y='reading score',hue='lunch')
```

```
Out[63]: <Axes: xlabel='race/ethnicity', ylabel='reading score'>
```



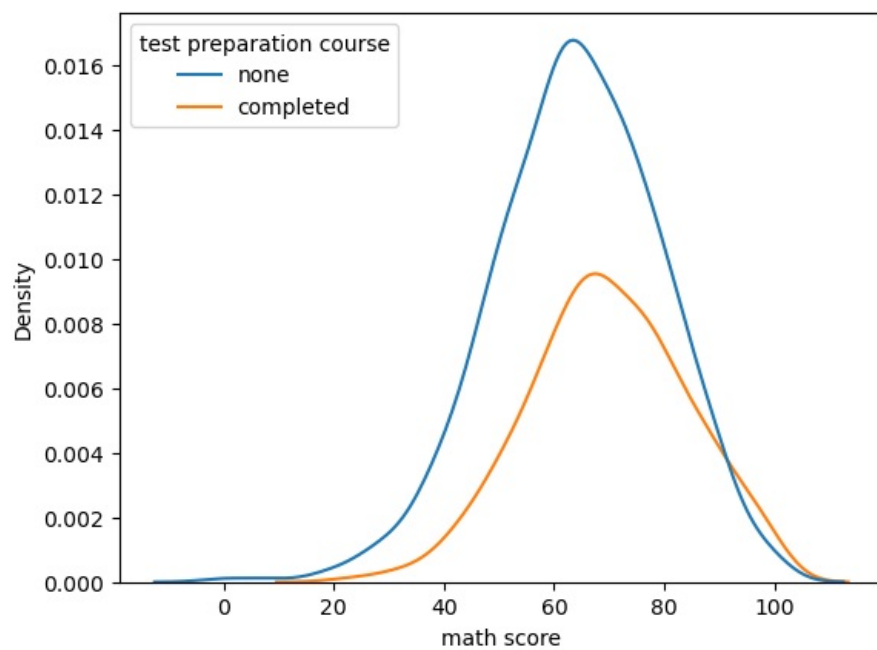
```
In [65]: sns.kdeplot(df,x='writing score',hue='parental level of education')
```

```
Out[65]: <Axes: xlabel='writing score', ylabel='Density'>
```



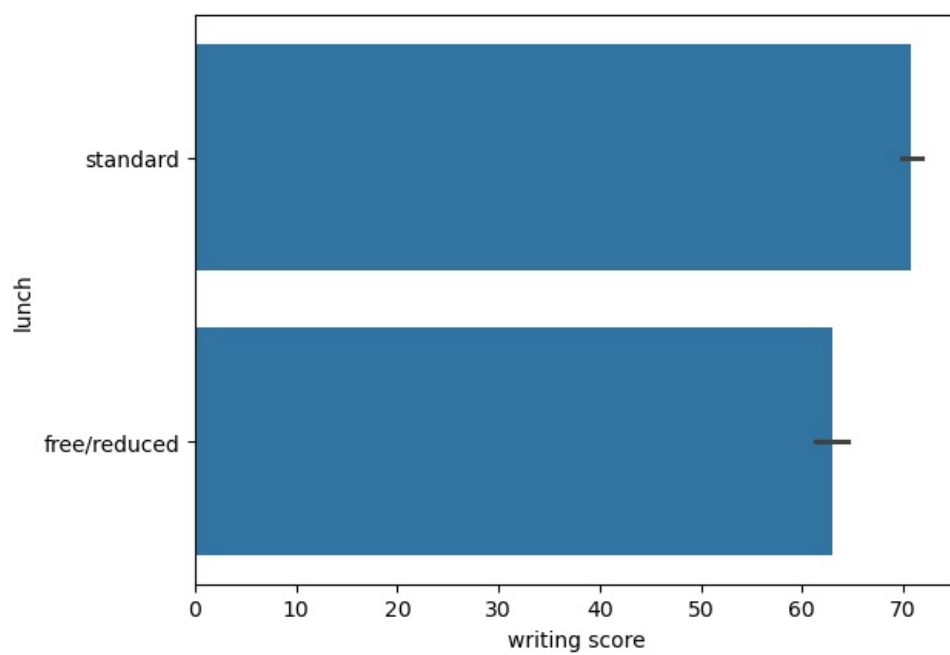
```
In [67]: sns.kdeplot(df,x='math score',hue='test preparation course')
```

```
Out[67]: <Axes: xlabel='math score', ylabel='Density'>
```

```
In [69]: sns.barplot(df,x='writing score',y='lunch')
```

```
Out[69]: <Axes: xlabel='writing score', ylabel='lunch'>
```



```
In [ ]:
```

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